Introduction to IoT and Embedded Systems

Module 1 Assignment

A great example of a device that has evolved from a non-IoT device to an IoT device is the **thermostat**.

Device Identified: Thermostat

Past (Non-IoT) Thermostat:

- **Functionality:**
  - Manually controlled.
  - Used mechanical or analog dials to set temperature.
  - Basic on/off or heating/cooling control.
- **Features:**
  - Simple interface.
  - No programmability.
  - Required physical presence to make adjustments.

Modern (IoT) Thermostat:

- **Functionality:**
  - Connected to the internet and cloud services.

    - Can be controlled remotely via smartphone or web apps.
    - Uses machine learning to auto-adjust temperature based on user habits.
    - Integrates with smart home ecosystems (like Alexa, Google Assistant).
- **Features:**
    - Remote access and monitoring.
    - Scheduling and automation.
    - Energy usage reports and recommendations.
    - Sensor integration (e.g., motion, humidity).
    - Over-the-air software updates.

Comparison:

Feature/Function	Non-IoT Thermostat	IoT Thermostat (e.g., Nest, Ecobee)
Control Method	Manual dial/switch	Mobile app / voice assistant
Remote Access	Not available	Available
Learning Capabilities	None	Learns user preferences
Energy Optimization	Manual only	Automated suggestions & reports
Integration with Other Devices	No interaction	Smart home compatibility
Internet Connectivity	Offline	Online & cloud-connected

Impact of IoT Capabilities:

The transformation of thermostats into IoT devices has led to more efficient energy use, convenience, and smarter living environments. It demonstrates how embedding connectivity and intelligence into everyday devices can significantly enhance their value and functionality.

Improvements in IoT Thermostat:

1. **Remote Control:** Users can adjust the temperature from anywhere using a smartphone or web app, providing convenience and flexibility.
2. **Energy Efficiency:** Smart thermostats learn user habits and optimize heating/cooling schedules, leading to potential energy and cost savings.
3. **Scheduling & Automation:** Automatically adjusts settings based on time of day, weather, or occupancy, reducing the need for manual changes.
4. **Integration:** Works with smart home systems and voice assistants like Google Assistant, Alexa, or Apple HomeKit.
5. **Usage Reports:** Provides insights into energy usage and suggestions for improving efficiency.
6. **User-Friendly Interface:** Touchscreen controls, app interfaces, and voice commands make operation easier for many users.

Diminishments (Disadvantages Compared to Non-IoT Version):

1. **Complexity:** Setup and operation may be overwhelming for non-technical users, especially the elderly or those unfamiliar with smartphones.
2. **Dependence on Internet:** Many features depend on a stable internet connection; without it, remote access and automation may not function.
3. **Privacy Concerns:** Data about household patterns and preferences is collected and stored by the manufacturer, raising privacy and data security issues.
4. **Cost:** Smart thermostats are significantly more expensive than basic manual thermostats.
5. **Software Bugs or Updates:** Like any smart device, performance may sometimes be affected by firmware bugs, app crashes, or failed updates.

Limitations of IoT Thermostat:

1. **Initial Configuration:** Requires compatible HVAC systems and correct installation, which may need professional help.

2. **Security Risks:** As with any internet-connected device, there is a risk of hacking if proper security measures are not in place.

3. **Dependence on Ecosystem:** Some devices work best only within a specific ecosystem (e.g., Google Nest with Google Home), limiting flexibility for mixed-device households.

4. **Power Requirements:** Unlike traditional thermostats powered by the HVAC system, some IoT models require batteries or continuous power, which can be a limitation in some setups.

Overall, while IoT thermostats offer many modern advantages in terms of convenience and efficiency, they also introduce complexities and dependencies that did not exist in their simpler, non-IoT counterparts.

Privacy Issues in IoT Thermostats (Not Present in Original Thermostats):

The original (non-IoT) thermostats were entirely offline, meaning they collected no user data and had zero privacy concerns. In contrast, modern IoT thermostats collect a variety of data, including:

- **User behavior and schedules** (e.g., when you're home or away).

- **Temperature preferences over time.**

- **Location data** through smartphone GPS or geofencing.

- **Integration data** from other smart devices in the home.

This data is often stored in the cloud and may be accessed by the manufacturer or third parties. Although this enables advanced features, it raises **privacy and surveillance concerns**, such as:

- Risk of unauthorized data access (hacking).

- Profiling and targeted advertising.

- Potential for data misuse if companies change privacy policies or are acquired.

Price Comparison:

Let’s compare the average price of a non-IoT thermostat in the early 2000s vs. a modern smart thermostat in 2025.

- **Original Non-IoT Thermostat (around 2000):**
  ~$30–$50 USD

- **Adjusted for Inflation (2025 dollars):**
  Using the U.S. Bureau of Labor Statistics CPI Inflation Calculator,
  $40 in 2000 ≈ **$70 in 2025**

- **Modern IoT Thermostat (2025):**
  ~$150–$250 USD (e.g., Google Nest, Ecobee, Honeywell T9)

Summary of Price Comparison:

Feature	Non-IoT Thermostat (2000)	IoT Thermostat (2025)
Average Price	$40	$150–$250
Adjusted for Inflation (2025)	~$70	$150–$250
Data Collection	None	Yes
Privacy Risks	None	Present

Conclusion:

IoT thermostats provide valuable modern features but introduce new concerns that did not exist with traditional thermostats, particularly regarding privacy and data security. Additionally, the cost of adoption has significantly increased, even when adjusted for inflation. This means users are paying more not just in dollars, but potentially in personal data as well.